



## **Definition of Triggers in SQL**

A **Trigger** is a special type of stored program in SQL that automatically executes when a specific event occurs on a table, such as:

- INSERT
- UPDATE
- DELETE

Triggers are mainly used for:

- Audit logging
- Data validation
- Automation
- Enforcing business rules

## **Definition of Events in SQL**

An **Event** is a scheduled database task that runs automatically at a specified time or interval.

Events are used for:

- Scheduled cleanup
- Automated reporting
- Archiving old data
- Background database maintenance

## **Short Notes and Explanation of Each Script**

### **SECTION 1: Introduction to Triggers**

```
SELECT *  
FROM job_roles;  
  
SELECT *  
FROM employees;
```

### **What It Does**

- Displays all records from:
  - job\_roles
  - employees



## Purpose

Helps inspect the tables before creating triggers.

## SECTION 2: Create Trigger

```
CREATE TRIGGER employee_after_insert  
AFTER INSERT ON employees
```

### What It Does

- Creates a trigger named employee\_after\_insert.
- Automatically executes after a new employee is inserted into the employees table.

## FOR EACH ROW

```
FOR EACH ROW
```

## Purpose

- Executes the trigger once for every inserted row.

## INSERT INTO employee\_audit\_log

```
INSERT INTO employee_audit_log
```

### What It Does

- Saves inserted employee details into an audit table.

## Purpose

- Tracks employee insert activities automatically.



## NEW Keyword

```
NEW.employee_id
```

### What It Does

- Refers to values from the newly inserted row.

### Example

- NEW.first\_name
- NEW.last\_name

## SECTION 3: Testing the Trigger

```
INSERT INTO employees (...)  
VALUES (...);
```

### What It Does

- Inserts a new employee record.

### Result

- Trigger automatically runs.
- Employee data is copied into employee\_audit\_log.

## View Audit Log

```
SELECT *  
FROM employee_audit_log;
```

### What It Does

- Displays audit records created by the trigger.



## SECTION 4: Delete Test Data

```
DELETE FROM employees  
WHERE employee_id = 101;
```

### What It Does

- Removes temporary testing data.

### Purpose

- Keeps database clean after trigger testing.

## SECTION 5: Introduction to Events

```
SHOW EVENTS;
```

### What It Does

- Displays all scheduled database events.

### Purpose

- Helps verify active automated tasks.

## SECTION 6: Create Automated Event

```
CREATE EVENT archive_old_employees
```

### What It Does

- Creates scheduled event named:  
archive\_old\_employees

### Event Schedule

```
ON SCHEDULE EVERY 1 DAY
```

### Purpose

- Runs the event automatically every day.



## Archive Logic

```
INSERT INTO archived_employees  
SELECT *  
FROM employees  
WHERE hire_date < '2015-01-01';
```

## What It Does

- Copies old employee records into archive table.

## Purpose

- Historical storage and data lifecycle management.

## SECTION 7: Automated Cleanup Event

```
CREATE EVENT cleanup_temp_reports
```

## What It Does

- Creates automatic cleanup process.

## Cleanup Logic

```
DELETE  
FROM temporary_reports  
WHERE created_at < NOW() - INTERVAL 7 DAY;
```

## What It Does

- Deletes temporary reports older than 7 days.

## Purpose

- Improves storage efficiency and database performance.



## SECTION 8: Verify Events

```
SHOW EVENTS;
```

### What It Does

- Displays active scheduled events.

### Purpose

- Confirms events are properly created and enabled.

### Key Concepts Used

| Concept      | Description                               |
|--------------|-------------------------------------------|
| TRIGGER      | Automatic SQL execution after table event |
| EVENT        | Scheduled SQL automation                  |
| AFTER INSERT | Executes after insertion                  |
| FOR EACH ROW | Runs trigger per inserted row             |
| NEW          | Accesses newly inserted values            |
| DELIMITER    | Changes SQL statement separator           |
| SHOW EVENTS  | Displays scheduled events                 |

### Benefits of Triggers

- Automatic auditing
- Data synchronization
- Security tracking
- Real-time automation
- Business rule enforcement



## Benefits of Events

- Scheduled automation
- Automatic cleanup
- Database maintenance
- Reporting automation
- Archiving old records

## Real-World Business Use Cases

### Triggers

Used for:

- Employee audit logging
- Banking transaction tracking
- Payroll automation
- Inventory updates
- Security monitoring

### Events

Used for:

- Daily backups
- Scheduled reporting
- ETL workflows
- Temporary data cleanup
- Archive management

## Overall Summary

This script demonstrates how MySQL automation works using:

### 1. Triggers

- React immediately to table changes.

### 2. Events

- Run scheduled database operations automatically.



Together, they help build:

- Automated enterprise systems
- Smart reporting solutions
- Self-maintaining databases
- Real-time business applications